WORKED SOLUTION

STEP III 2014, Question 1

Pure • Logarithmic Series Expansion

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q1

The key idea

The cubic $1+qx^2+rx^3$ factorises as $(1+ax)(1+bx)(1+cx)$, so its logarithm splits into three logarithms, each with a series we know. This gives the coefficient of x^n in terms of a, b, c. We can also expand the same logarithm directly as $\ln(1+u)$ with $u=qx^2+rx^3$, and that gives the coefficients in terms of q and r. Comparing the two expansions turns facts about q and r into identities between the S_n.

Solution

Multiplying out,

$$(1+ax)(1+bx)(1+cx) = 1+(a+b+c)x+(ab+bc+ca)x^2+abc\,x^3.$$

Since $a+b+c=0$, the x term vanishes. Two polynomials that agree for all real x have equal coefficients, so comparing the coefficients of x^2 and x^3 with $1+qx^2+rx^3$ gives

$$q = bc+ca+ab, \qquad r = abc,$$

as required.

Part (i)

For $|x|$ small enough (specifically $|x|<1/\max(|a|,|b|,|c|)$, or any x if $a=b=c=0$), each factor is positive and each logarithm has a convergent series. Using $\ln(1+t)=\sum_{n\geq 1}(-1)^{n+1}t^n/n$ for $|t|<1$,

$$\begin{aligned}\ln(1+qx^2+rx^3) &= \ln(1+ax)+\ln(1+bx)+\ln(1+cx)\\ &= \sum_{n\geq 1}(-1)^{n+1}\frac{a^nx^n}{n}+\sum_{n\geq 1}(-1)^{n+1}\frac{b^nx^n}{n}+\sum_{n\geq 1}(-1)^{n+1}\frac{c^nx^n}{n}\\ &= \sum_{n\geq 1}(-1)^{n+1}\frac{a^n+b^n+c^n}{n}\,x^n.\end{aligned}$$

So the coefficient of x^n is $(-1)^{n+1}S_n$, as required. (For $n = 1$ this is $S_1 = a + b + c = 0$, which matches the absence of an x term.)

Part (ii)

Now expand the same function a second way. Put $u = qx^2 + rx^3$, so

$$\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \frac{u^4}{4} + \cdots.$$

Every term of u has degree at least 2, so u^k involves only powers x^{2k} up to x^{3k}. To find the coefficient of x^n we only need the u^k with $2k \le n \le 3k$:

$$u = qx^2 + rx^3,$$
$$u^2 = q^2x^4 + 2qr\,x^5 + r^2x^6.$$

Hence:

- x^2: only u contributes, giving q;
- x^3: only u contributes, giving r;
- x^5: only $-\frac{1}{2}u^2$ contributes, giving $-\frac{1}{2}(2qr) = -qr$.

Comparing with part (i), where the coefficients are $-S_2$, S_3 and S_5,

$$S_2 = -q, \qquad S_3 = r, \qquad S_5 = -qr.$$

Therefore $S_2S_3 = (-q)(r) = -qr = S_5$, as required.

Ask yourself Why are we allowed to compare coefficients of two different series for the same function?

Both expansions are power series in x that converge to $\ln(1 + qx^2 + rx^3)$ on some interval around 0. A power series that converges near 0 has unique coefficients (they are the derivatives at 0 divided by $n!$, as in Maclaurin's theorem), so the coefficients must agree.

Part (iii)

For x^7 we need $2k \le 7 \le 3k$, which forces $k = 3$. Expanding $u^3 = (qx^2 + rx^3)^3$ by the binomial theorem,

$$u^3 = q^3x^6 + 3q^2r\,x^7 + 3qr^2x^8 + r^3x^9,$$

so the coefficient of x^7 in $\ln(1+u)$ is $\frac{1}{3} \cdot 3q^2r = q^2r$. From part (i) it is $(-1)^8S_7 = S_7$, so $S_7 = q^2r$. Using part (ii),

$$S_2S_5 = (-q)(-qr) = q^2r = S_7,$$

as required.

Part (iv)

For x^9 we need $2k \leq 9 \leq 3k$, so $k = 3$ or $k = 4$.

- From $\frac{1}{3}u^3$: the x^9 term of u^3 is r^3x^9, giving $\frac{1}{3}r^3$.
- From $-\frac{1}{4}u^4$: in $(qx^2 + rx^3)^4$, the x^9 term needs three factors of qx^2 and one of rx^3, giving $\binom{4}{1}q^3r\,x^9 = 4q^3r\,x^9$. This contributes $-q^3r$.

So the coefficient of x^9 is $\frac{1}{3}r^3 - q^3r$, and by part (i) this equals $(-1)^{10}S_9 = S_9$. On the other hand,

$$S_2S_7 = (-q)(q^2r) = -q^3r.$$

Hence

$$S_9 - S_2S_7 = \tfrac{1}{3}r^3 = \tfrac{1}{3}(abc)^3,$$

and the claim $S_2S_7 = S_9$ holds exactly when $abc = 0$. It is false in general.

For a concrete counterexample take $a = b = 1$, $c = -2$ (so $a + b + c = 0$ and $abc = -2 \neq 0$). Then

$$S_2 = \frac{1+1+4}{2} = 3, \qquad S_7 = \frac{1+1-128}{7} = -18, \qquad S_9 = \frac{1+1-512}{9} = -\frac{170}{3},$$

so $S_2S_7 = -54 \neq -\frac{170}{3} = S_9$. The claim is false.

Ask yourself Parts (ii) and (iii) worked, so why does the pattern break at x^9?

For x^5 and x^7 only one power of u contributes, so each coefficient is a single product of qs and rs that happens to factorise as the right product of earlier Ss. At x^9 two powers of u contribute, and the extra r^3 term from u^3 spoils the factorisation.

Checks

- With $a = b = 1$, $c = -2$ we have $q = -3$ and $r = -2$. Then $S_2 = 3 = -q$, $S_3 = (1 + 1 - 8)/3 = -2 = r$ and $S_5 = (1 + 1 - 32)/5 = -6 = -qr$, so $S_2S_3 = S_5$. Also $S_7 = -18 = q^2r$, and $S_9 - S_2S_7 = -\frac{170}{3} + 54 = -\frac{8}{3} = \frac{1}{3}r^3$, exactly as predicted.
- If $c = 0$ (so $b = -a$), all odd S_n vanish and both sides of the claim in (iv) are 0, consistent with the claim holding when $abc = 0$.
- The identities in (ii)–(iv) and the formula for $S_9 - S_2S_7$ were also confirmed by computer, in exact arithmetic, for 200 random rational choices of a and b.

Common slips

- Missing the u^4 contribution to x^9: it is easy to stop at u^3, giving the wrong coefficient $\frac{1}{3}r^3$.
- Sign errors from $(-1)^{n+1}$: the coefficient of x^2 is $-S_2$, so $S_2 = -q$, not q.

- Forgetting the binomial coefficient: the x^7 term of u^3 is $3q^2r$, not q^2r.

Beyond STEP

The S_n are (up to the factor $1/n$) the *power sums* of the roots of a cubic, and q, r are its *elementary symmetric polynomials*. Taking logarithms and comparing coefficients is a generating-function proof of Newton's identities, which express every power sum in terms of the elementary symmetric polynomials; the theorem behind it is that every symmetric polynomial can be written in terms of the elementary ones. You can read about symmetric polynomials and roots of polynomials in Allenby, *Rings, Fields and Groups*, and find more STEP-style practice with series in Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.